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MAR_APR 2026 SUMMER EXAMINATION

1154 B.Tech. CBCS

Sub. Name: Switchgear and Protection

Sub. Code: 67517/83779/83974

Day and Date: Friday ,15-05-2026 **Total Marks: 70**

Time: 10:30 AM To 01:00 PM

Instructions: **1. All questions are compulsory**
2. Draw neat labelled diagrams wherever necessary
3. Figures to the right indicate full marks

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Q1) Answer any two of the following questions. **[12]**

a. Explain High resistance interruption in detail. **[6]**

b. Explain Air break circuit breaker in detail. **[6]**

c. Write note on HRC fuse, fuse characteristics, application. **[6]**

Q2) Answer any two of the following questions. **[12]**

a. Explain Induction cup type relay in detail. **[6]**

b. Explain attracted armature type relay in detail. **[6]**

c. Which are the desirable qualities of protective relays explain in detail. **[6]**

Q3) Answer any two of the following questions. **[12]**

a. Explain directional relay in detail. **[6]**

b. Write note on Plug setting and Time setting in detail. **[6]**

c. Explain microprocessor based over voltage relay. **[6]**

Q4) Answer any two of the following questions. **[12]**

a. Explain carrier current protection scheme for line protection. **[6]**

b. Explain circulating current & opposed voltage principle of differential relay. **[6]**

c. Explain the transmitting unit used in carrier aided protection. **[6]**

Q5) Answer any two of the following questions. **[12]**

a. Explain problems associated with percentage differential protection. **[6]**

b. Write a note on, **[6]**

(i) Magnetic current inrush

(ii) Frame leakage protection

c. Explain percentage differential protection for transformer. **[6]**

Q6) Answer any two of the following questions. **[10]**

a. What are the causes of over-voltage? How the lightning arresters provide protection against them? **[5]**

b. Explain stator earth fault & phase fault protection of alternator. **[5]**

c. Explain relay setting for 3 zone distance protection **[5]**

End Of Question Paper

Important Note for Chief Exam Officer / SRPD Coordinator / Sr Supervisor/ Student -
This Question Paper may be distributed for following Subjects as common code.
सदरची प्रश्नपत्रिका खालील विषयांकरिता वितरित करता येईल.

1] (101) Bachelor of Engineering (67517) Advanced Switchgear and Protection Part 4 SEM 7

2] (101) Bachelor of Engineering (83974) Switchgear and Protection Part 4 SEM 7

3] (1154) B.Tech. CBCS (83779) Switchgear and Protection Part 4 SEM 7

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